1: Common Network Issues Theory :Common hardware and software issues.Practical :Identify faulty devices in a simulated network.Task: Create a troubleshooting checklist.

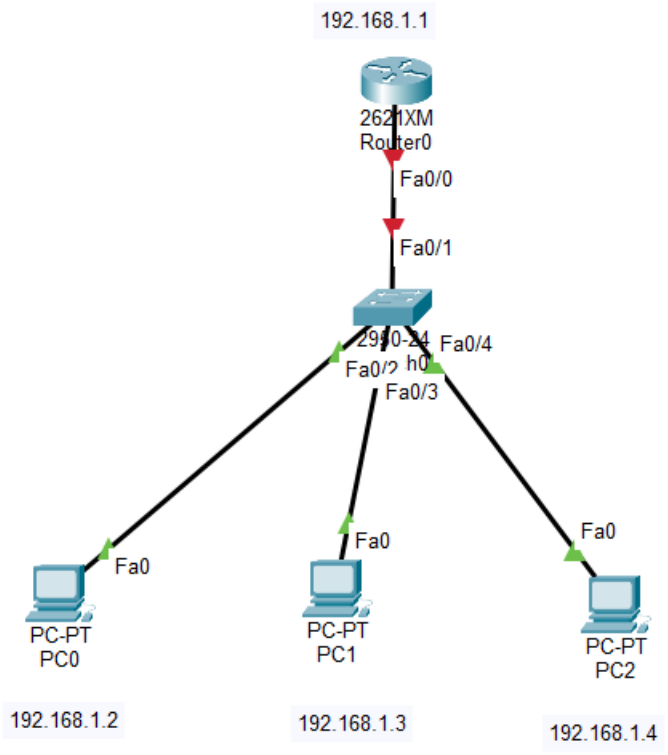
Task 1.

List 5 common hardware issues that can cause network problems in a typical office setup, and for each, describe one symptom that helps identify it.

No.	Hardware issue	Checklist
1	Damaged Ethernet cable	Network connection keeps disconnecting or shows “Network cable unplugged.”
2	Faulty switch port	Devices work when connected to another switch port but not the faulty port.
3	Faulty NIC (Network Interface Card)	The computer does not detect an Ethernet connection even with a known-good cable.
4	Router failure/overheating	Multiple devices lose Internet access, and the router may restart or become unresponsive.
5	Loose power cable/power adapter	Router or switch suddenly powers off or repeatedly restarts.

Task 2.

Open Cisco Packet Tracer or any network simulator, set up a small network with 3 PCs, 1 switch, and 1 router, and intentionally disconnect one device. Use the simulator's troubleshooting tools to identify which device is faulty and document the steps you took.




```
Cisco Packet Tracer PC Command Line 1.0
C:\>ping 192.168.1.3

Pinging 192.168.1.3 with 32 bytes of data:

Reply from 192.168.1.3: bytes=32 time<1ms TTL=128
Reply from 192.168.1.3: bytes=32 time<1ms TTL=128
Reply from 192.168.1.3: bytes=32 time<1ms TTL=128
Reply from 192.168.1.3: bytes=32 time<1ms TTL=128

Ping statistics for 192.168.1.3:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 0ms, Average = 0ms

C:\>ping 192.168.1.4

Pinging 192.168.1.4 with 32 bytes of data:

Reply from 192.168.1.4: bytes=32 time<1ms TTL=128
Reply from 192.168.1.4: bytes=32 time<1ms TTL=128
Reply from 192.168.1.4: bytes=32 time<1ms TTL=128
Reply from 192.168.1.4: bytes=32 time<1ms TTL=128

Ping statistics for 192.168.1.4:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 0ms, Average = 0ms

C:\>ping 192.168.1.1

Pinging 192.168.1.1 with 32 bytes of data:

Request timed out.
Request timed out.
Request timed out.
Request timed out.

Ping statistics for 192.168.1.1:
    Packets: Sent = 4, Received = 0, Lost = 4 (100% loss),

C:\>
```

Intentionally disconnect a device

1. Select the Delete tool (red X) in Packet Tracer.
2. Click the cable connecting PC2 to Switch0.
3. PC2 is now disconnected from the network**.**

Step: Identify the fault

1. Observe the link between PC2 and Switch0. It should show a disconnected link.
2. On PC0, run ping 192.168.1.4.
3. The ping should fail because PC2 is disconnected.
4. Check PC2's connection to confirm that its cable is missing.
5. Reconnect PC2 using a Copper Straight-Through cable.
6. Repeat the ping test. It should succeed after the link becomes active.

Troubleshooting record

Test	Observation
PC0 → PC1 ping	Successful
PC0 → PC2 ping	Failed after disconnection
Visual inspection	PC2 cable disconnected
After reconnecting	PC0 → PC2 ping succeeds

Conclusion: The fault was identified as a disconnected network cable between PC2 and the switch.

Task 3.

Create a troubleshooting checklist for diagnosing slow internet issues in a home WiFi network, including at least 8 steps covering both hardware and software checks.

Hint: Think about checking cables, router status lights, device settings, and running ping or speed tests.

3. Troubleshooting checklist for slow home Wi-Fi

Use this checklist from basic physical checks to software/network checks.

Step 1 — Check router power

Make sure the router is powered on properly and check its status LEDs.

Step 2 — Check Ethernet/Internet cable

If the router connects to an ISP modem/ONT, check that the Ethernet cable is properly connected.

Try another cable if you suspect cable damage.

Step 3 — Check Wi-Fi signal strength

Move the laptop/phone closer to the router.

If the speed improves significantly, weak Wi-Fi signal or interference may be the problem.

Step 4 — Restart the router

Power off the router, wait approximately 30 seconds, and turn it back on.

Wait until the Internet/WAN indicator returns to normal.

tep 5 — Check how many devices are connected

Too many devices may consume available bandwidth.

Check the router's connected-device list and disconnect devices that are not needed.

Step 6 — Check the device's Wi-Fi settings

On the laptop:

- Make sure Wi-Fi is enabled.
- Disconnect and reconnect to the Wi-Fi network.
- Verify that Airplane Mode is off.
- Forget and reconnect to the Wi-Fi network if necessary.

Step 7 — Test Internet connectivity using ping

Open Command Prompt:

ping 8.8.8.8

If the ping shows high latency or packet loss, there may be a connectivity problem.

You can also test DNS:

ping google.com

Step 8 — Run an Internet speed test

Use a reputable speed-test service and compare the result with the Internet plan's expected speed.

If possible, test using Ethernet and Wi-Fi.

Step 9 — Check background applications

Programs such as:

- Windows Update
- cloud synchronization
- downloads
- streaming applications

may consume bandwidth.

Check **Task Manager → Performance/Processes → Network**.

Step 10 — Check router placement/interference

Keep the router away from:

- Thick walls
- Microwaves
- Large metal objects
- Other sources of radio interference

For 2.4 GHz Wi-Fi, nearby networks can also cause congestion.

Task 4.

You receive a complaint that a user cannot access Instagram and YouTube on their laptop, but WhatsApp Web works fine. Write down the first 5 troubleshooting steps you would perform to identify if the issue is with hardware, software, or network configuration.

Because **WhatsApp Web works**, the laptop has at least some working network connectivity. I would troubleshoot in this order:

Step 1 — Test basic Internet connectivity

Open Command Prompt:

ping 8.8.8.8

If this fails, investigate the network connection, Wi-Fi, router, or ISP.

Step 2 — Test DNS resolution

Run:

nslookup youtube.com

nslookup instagram.com

If IP addresses are not returned, the problem could be **DNS configuration**.

You can also test:

ping google.com

This helps distinguish basic IP connectivity from DNS problems.

Step 3 — Try another browser

For example:

- Chrome
- Edge
- Firefox

If Instagram and YouTube work in another browser, the problem may be related to the original browser's **cache, extensions, cookies, or settings**.

Step 4 — Check browser cache, extensions, and proxy settings

Temporarily disable unnecessary browser extensions and clear cached data.

Also check:

Windows Settings → Network & Internet → Proxy

Make sure an unwanted manual proxy is not configured.

Step 5 — Check firewall/security software and network configuration

Check whether Windows Firewall or third-party security software is blocking the websites.

Also check:

ipconfig /all

Look at:

- IP address
- Default gateway
- DNS servers
- Network adapter status

Then test the gateway:

ping <default-gateway>

For example:

ping 192.168.1.1

Test result	**Likely area**
Cannot ping gateway	Wi-Fi/NIC/router/hardware
Gateway works, 8.8.8.8 fails	Internet/ISP/router
8.8.8.8 works but domain lookup fails	DNS
DNS works but websites fail in one browser	Browser/software
Websites fail across browsers but WhatsApp works	Firewall/proxy/DNS/network filtering
Websites work on another device using same Wi-Fi	Laptop-specific problem